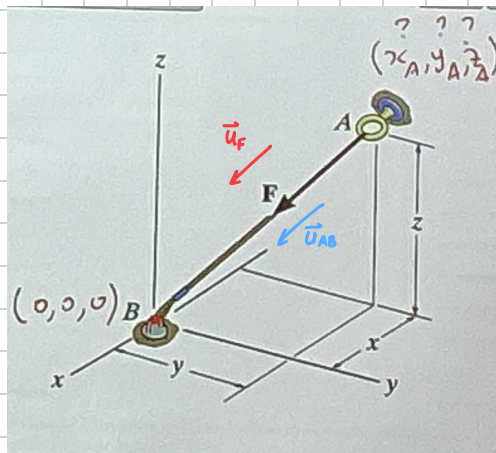


Week 5



AB is 9m long

$$F = \{ 350i - 250j - 450k \} N$$

find (x_A, y_A, z_A)

$$|\vec{r}_{AB}| = 9.0m$$

$$① \vec{u}_F = \frac{\vec{F}}{|\vec{F}|}$$

↓
direction of force F

$$② \vec{u}_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} ?$$

$$③ \vec{u}_F = \vec{u}_{AB} \rightarrow \text{Solve for } x_A, y_A, z_A$$

$$① \quad \vec{F} = \{350\mathbf{i} - 250\mathbf{j} - 450\mathbf{k}\} \text{ N}$$

$$|\vec{F}| = \sqrt{350^2 + 250^2 + 450^2} = 622.5 \text{ N}$$

$$\vec{U}_F = \frac{\vec{F}}{|\vec{F}|} = \{0.5623\mathbf{i} - 0.4016\mathbf{j} - 0.7229\mathbf{k}\}$$

$$|\vec{r}_{AB}| = 9.0 \text{ m}$$

$$\vec{r}_{AB} = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}$$

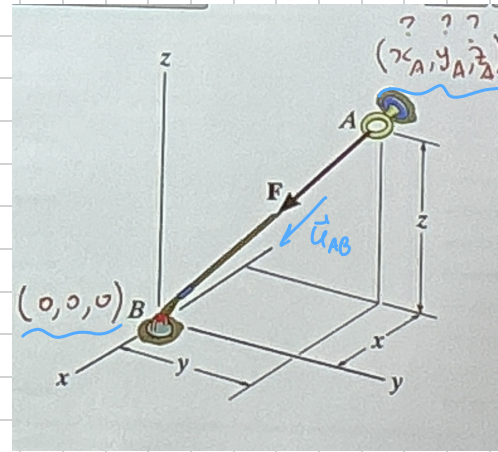
$$= \{-x_A\mathbf{i} - y_A\mathbf{j} - z_A\mathbf{k}\} \text{ m}$$

$$\vec{U}_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|}$$

↳ direction of force

$$\vec{U}_{AB} = \left\{ \frac{-x_A}{9.0}\mathbf{i} - \frac{y_A}{9.0}\mathbf{j} - \frac{z_A}{9.0}\mathbf{k} \right\}$$

$$\vec{U}_F = \vec{U}_{AB}$$



$$\vec{U}_F = 0.5623\mathbf{i} - 0.4016\mathbf{j} - 0.7229\mathbf{k} \rightarrow ①$$

$$\vec{U}_{AB} = \frac{-x_A}{9}\mathbf{i} - \frac{y_A}{9}\mathbf{j} - \frac{z_A}{9}\mathbf{k} \rightarrow ②$$

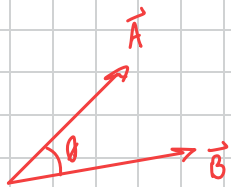
$$-\frac{x_A}{9} = 0.5623 \rightarrow x_A = -5.06 \text{ m} \rightarrow ①$$

$$-\frac{y_A}{9} = -0.4016 \rightarrow y_A = 3.61 \text{ m} \rightarrow ②$$

$$-\frac{z_A}{9} = -0.7229 \rightarrow z_A = 6.51 \text{ m} \rightarrow ③$$

Quiz One! vectors part 4

Dot product



finding Angle from two vectors

Written as $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$

↳ scalar quantity output

Laws:

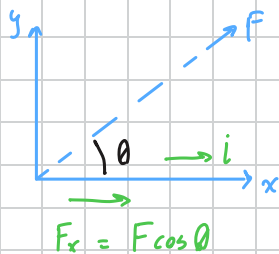
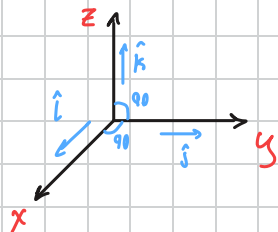
① $A \cdot B = B \cdot A$

② $a(A \cdot B) = (aA) \cdot B = A \cdot aB$

③ $A \cdot (B + D) = A \cdot B + A \cdot D$

$$\hat{i} \cdot \hat{i} = (1)(1) \cos 0^\circ = \underline{\underline{1}} \quad \hat{i} \cdot \hat{j} = (1)(1) \cos 90^\circ = \underline{\underline{0}}$$

$$\hat{i} \cdot \hat{k} = (1)(1) \cos 90^\circ = \underline{\underline{0}}$$



$$\vec{F} \cdot \hat{i} = |\vec{F}| |\hat{i}| \cos \theta$$

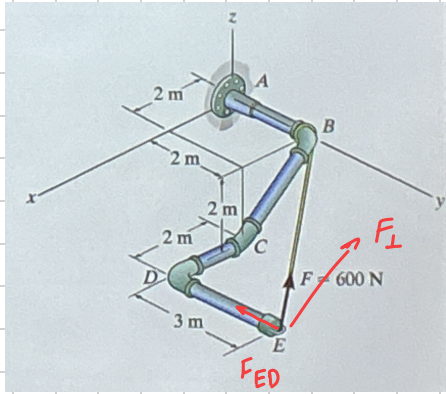
$$F_x = F(1.0) \cos \theta$$

dot product to itself i.e. $\hat{k} \cdot \hat{k} = 1$
to others = 0

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= (A_x B_x + A_y B_y + A_z B_z) \rightarrow \text{Scalar} \end{aligned}$$

$$\begin{aligned} &A_x B_x \hat{i} \cdot \hat{i} + A_x B_y \hat{i} \cdot \hat{j} + A_x B_z \hat{i} \cdot \hat{k} \\ &+ A_y B_x \hat{j} \cdot \hat{i} + A_y B_y \hat{j} \cdot \hat{j} + A_y B_z \hat{j} \cdot \hat{k} \\ &+ A_z B_x \hat{k} \cdot \hat{i} + A_z B_y \hat{k} \cdot \hat{j} + A_z B_z \hat{k} \cdot \hat{k} \end{aligned}$$

$$\theta = \cos^{-1} \left(\frac{A_x B_x + A_y B_y + A_z B_z}{AB} \right)$$



$$A_{||} = A \cdot u = A \cos \theta$$